

Notes on Lance Taylor (2009) *Reconstructing Macroeconomics*
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SOCIAL ACCOUNTING MATRIX

Households of workers can be quite different from the households of rentiers or capital owners. Workers mostly receive incomes as wages and government transfers. Rentier households may receive incomes as interest, dividend and rent. At this stage, we do not want to make the distinction between the two classes of households. Let us add up all the household incomes: Wages + Transfers + Interest on deposits + Dividend on equity.

(We can ignore rent too. One group of households receive rent, while another group pay rent. In the aggregate, they cancel out)

Households receive incomes from wages paid to them (wbX), transfers from the government (Q_H), interest (at the rate i) received on workers' bank deposits (M_H), and dividend payments ($\delta P \cdot V$).

Note: w is wages per worker; X is output; b is the Number of workers (L) per unit of output (X) or $b = L/X$. wbX is incomes in the form of wages received by all worker households combined.

V is the volume of equity issued by business firms, all of which (we assume) are held by rentiers, P_v is the price of equity, δ is dividend payout rate.

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Assume you hold 50 shares of a company, with each share selling at a price (average for, say, a year) of Rs.1000. The value of your shares (which is your wealth) is Rs.50,000. If shareholders of this company receive a dividend of, on average, 6%, your annual income from your shares will be Rs. Rs.3000.

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Households spend their incomes on consumption (PC_h), and payment of taxes by households (T_h), and the rest of their incomes are saved (S_h)

Workers receive their incomes from: $Y_h = \text{Wages} + \text{Transfers} + \text{Interest on deposits} + \text{Dividend on equity}$

$$Y_h = w_b X + Q_h + iM_h + \delta P_v V$$

What households spend their incomes on: $Y_h = \text{Consumption} + \text{Taxes} + \text{Savings}$

$$= PC_h + T_h + S_h$$

Businesses receive their incomes from: $Y_B = \text{Profits } (\Pi PX)$

Π is profit share of output, P is prices (or the price deflator); X is real output (tonnes of steel); PX is output in nominal values (Rupees of dollars worth of output).

Here we are treating business as an entity separate from other owners of capital including (in India) some families that run big businesses). Businesses make new investments (in factories, and so on) by raising financial resources from the market (bank loans or equity). The profits they earn are used to pay interest and dividend on the money they raised and to pay taxes. In addition, a part of the profits may be saved (as retained earnings).

What businesses spend their incomes on:

$$\begin{aligned} Y_B &= \text{Interest on Loans (to business)} + \text{Dividend on Shares} + \text{Taxes (business)} \\ &+ \text{Savings (business)} \\ &= \delta P_v V + i L_B + T_B + S_B \end{aligned}$$

Note: Profits are what the businesses get to keep after making the payments on inputs and wages.

For firms, the finance for running their businesses comes from the equity raised from the market, loans from domestic banks and foreign financial

institutions, and their retained earnings (savings used for future business operations or expansion).

The profit incomes the firms receive are deployed to pay:

Dividends for equity holders (of the shares issued by the firms), interest on the loans they have taken from domestic and foreign sources, and taxes. The rest of the profits are saved.

Governments receive their incomes from: $Y_g = T_x + T_H + T_B$

What governments spend their incomes on

$$Y_g = PG + Q_H + iL_g + S_g$$

The domestic interest rate is 'i'

We are not making a distinction between lending rates and deposit rates at this stage.

L_B and L_g are the loans given by the domestic banking system to businesses and the government, respectively

T_x is taxes minus subsidies on production (indirect taxes).

T_w , T_r , and T_b are taxes received from workers, rentiers and businesses.

Summarizing all the above, this is what we get:

	Where incomes come from:	Where incomes go to:
Households	Wages + Transfers (from govt)+Interest on deposits + Dividend on shares	Consumption + Taxes (households) + Savings (households)
Business	Profits	Interest on Loans (to business) + Dividend on Shares + Taxes (business) + Savings (business)
Government	Taxes = Indirect taxes (on production) + Taxes (households)+ Taxes (business)	Govt. Expenditure + Transfers (govt.) + Interest on loans (to government) + Savings (government)
Banks	Interest received on Loans (to government) + Interest on Loans (to business)	Interest to be paid on Deposits (from households)

Items in black font appear on both sides and they cancel out. So what we get is the following:

$$\text{Wages} + \text{Profits} + \text{Indirect taxes} = \text{Consumption} + \text{Govt. Expenditure} + \text{Savings (households)} + \text{Savings (business)} + \text{Savings (government)}$$

Savings is channelled (through banks and financial institutions) to Investment

$$\text{Wages} + \text{Profits} + \text{Indirect taxes} = \text{Consumption} + \text{Investment} + \text{Govt. Expenditure}$$

Writing the above using algebra:

	Where incomes come from:	Where incomes go to:
Households	$wbX + Q_H + iM + \delta P_v V$	$PC + T_H + S_H$
Business	ΠPX	$iL_B + \delta P_v V + T_B + S_B$
Government	Taxes = $T_x + T_H + T_B$	$PG + Q_H + iL_G + S_G$
Banks	$iL_G + iL_B$	iM

Summing up the left and right hand sides and leaving out items that appear on both sides, this is what we get:

$$wbX + \Pi PX + T_x = PC + PG + S_H + S_B + S_G$$

$$S_H + S_B + S_G = PI$$

What the banks do is summarized below: Mobilize deposits (and paying interest to savers) and use the deposits to give loans (to business and government)

$$iM = iL_G + iL_B$$

IN AN OPEN ECONOMY:

So far, we have considered a closed economy. Now in an open economy, we have to consider the economic transactions our country have with the rest of the world.

Where 'foreign' (all external entities that are engaged in economic transactions with the domestic economy)

receive their incomes from: $Y_f = exX + i^* e Z_g + i^* e Z_b$

What foreign spend their incomes on $Y_f = PE + i^* e R + S_f$

Foreigners receive income from the payments we (the home country) make for our imports (exX) and from the interest paid by businesses and government (in the home country) on the foreign loans they took.

Foreigners spend the income they receive (from us) to pay for the goods and services we export to them (PE) and also to pay interest on the foreign exchange reserves that are held by us (the home country) (this is: $i^* e R$)

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Note: Our foreign exchange reserves are, in effect, the loans we have given to the foreign country.

a is imports (or inputs sourced from abroad) as a share of output (X); e is the exchange rate. aX is the value of imports in, say, dollars, and eaX is the value of imports in Rupees.

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	Where incomes come from:	Where incomes go to:
Households	Wages + Transfers (from govt)+Interest on deposits + Dividend on shares	Consumption + Taxes (households) + Savings (households)
Business	Profits	Taxes + Interest on loans (to business) + Dividend on Shares + Interest on foreign loans (to business) + Savings (business)
Government	Taxes = Indirect taxes (on production) + Direct taxes (households)+ Direct taxes (business)	Govt. Expenditure + Transfers (govt.) + Interest on loans (to government) + Interest on foreign loans (to government) + Savings (government)
Banks	Interest received on Loans (to government) + Interest on Loans (to business)	Interest to be paid on Deposits (from households)
Foreign	Imports (by the home country) + Interest received on foreign loans ((to government) + Interest received on foreign loans (to business)	Exports (by the home country) + Interest received on foreign exchange reserves (loans to foreign) + Savings (foreign)

The above relations written using algebra:

	Where incomes come from:	Where incomes go to:
Households	$wbX + Q_H + iM + \delta P_v V$	$PC + T_H + S_H$
Business	ΠPX	$iL_B + \delta P_v V + i^* e Z^*_B + T_B + S_B$
Government	Taxes = $T_x + T_H + T_B$	$PG + Q_H + iL_G + i^* e Z^*_G + S_G$
Banks	$iL_G + iL_B + i^* e R^*$	iM
Foreign	$eaX + i^* e Z^*_G + i^* e Z^*_B$	$PE + i^* e R^* + S_F$

$$wbX + \Pi PX + T_x = PC + PG + (PE - eaX) + S_H + S_B + S_G + S_F$$

$$\text{Wages} + \text{Profits} + \text{Indirect taxes} = \text{Consumption} + \text{Govt. Expenditure} + (\text{Exports} - \text{Imports}) + \text{Savings (households)} \\ \text{Savings (business)} \text{ Savings (government)}$$

STOCKS AND FLOWS

Social classes	Liabilities	Assets
1. Households	Ω_h	M_h
2. Business	$L_b + P_b V + \Omega_b$	qPK
4. Government	L_g	Δ
5. Banks	M	L

Social classes	Liabilities	Assets
1. Households	Ω_h	M_h
2. Business	$L_b + eZ_b^* + P_b V + \Omega_b$	qPK
4. Government	$L_g + eZ_g^*$	Δ
5. Banks	M	$L + eR^*$
6. Foreign	$eR^* + \Omega$	eZ^*